

Literature Review on Drug Development

Pembrolizumab (Keytruda)



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Introduction

The development of immunotherapy has significantly changed the treatment of cancer. Traditional therapies such as chemotherapy and radiotherapy are effective in many patients, but they often affect healthy cells along with cancer cells. In recent years, researchers have focused on treatments that enhance the body's own immune response against cancer. One of the most successful examples of this approach is pembrolizumab, marketed under the brand name Keytruda.

Pembrolizumab is a humanized monoclonal antibody that targets the programmed cell death-1 (PD-1) receptor on T-cells. By blocking this receptor, the drug restores the ability of the immune system to recognize and attack cancer cells. Since its approval, pembrolizumab has become one of the most widely used immunotherapy drugs and has demonstrated remarkable clinical benefits across multiple cancer types.

Section 1: Drug Discovery Process

Pembrolizumab is a humanized monoclonal antibody developed to target the programmed cell death-1 (PD-1) receptor found on activated T-cells. The development of this drug was based on decades of research in cancer immunology and immune checkpoint pathways. The foundation of this discovery was laid by Dr. Tasuku Honjo, who identified the PD-1 receptor in 1992. His work, along with Dr. James Allison's research on immune checkpoints, was recognized with the Nobel Prize in Physiology or Medicine in 2018.

Researchers later discovered that many cancer cells use programmed death ligand-1 (PD-L1) proteins to bind with PD-1 receptors on T-cells, thereby suppressing the immune response and escaping immune surveillance. Under normal physiological conditions, the PD-1/PD-L1 pathway helps regulate immune activity and prevents excessive immune reactions. However, cancer cells exploit this mechanism to avoid destruction by the immune system.

To overcome this challenge, Merck developed pembrolizumab, which was initially known as MK-3475 during its research and development phase. Pembrolizumab binds specifically to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2. As a result, T-cells remain active and are able to recognize and destroy cancer cells more effectively.

The development of pembrolizumab represented a major breakthrough in cancer immunotherapy because it enhanced the body's natural immune response rather than directly attacking tumor cells. This innovative approach opened new possibilities for treating advanced cancers and has significantly improved outcomes for many patients worldwide.

Section 2: Clinical Trial Phases

The clinical development of pembrolizumab followed the standard sequence of Phase I, Phase II, Phase III, and post-marketing Phase IV studies. Each phase played an important role in establishing the safety, efficacy, and long-term clinical value of the drug.

Clinical trials generally begin with Phase I studies, where a new drug is tested in a small group of patients to evaluate safety, tolerability, and appropriate dosage. For pembrolizumab, the KEYNOTE-001 study served as the first major Phase I clinical trial. This study enrolled patients with advanced cancers, particularly melanoma, and was designed to determine the optimal dose while assessing the drug's safety profile. The results demonstrated acceptable safety and encouraging antitumor activity, providing the basis for further clinical development.

After safety was established, Phase II studies focused on evaluating the therapeutic effectiveness of pembrolizumab in different cancer types. Positive outcomes from these studies supported progression to larger confirmatory trials.

One of the most important Phase III studies was KEYNOTE-006, conducted in patients with advanced melanoma. In this trial, pembrolizumab was compared with ipilimumab, a widely used immunotherapy drug. The study demonstrated superior overall survival and progression-free survival with pembrolizumab, making it a preferred treatment option for melanoma patients.

Another landmark Phase III trial was KEYNOTE-024, which involved patients with advanced non-small cell lung cancer (NSCLC) who showed high PD-L1 expression. Pembrolizumab was compared with platinum-based chemotherapy and demonstrated significantly better progression-free survival and overall survival. These findings established pembrolizumab as an important first-line treatment option for eligible NSCLC patients.

Based on the evidence generated from these clinical trials, the United States Food and Drug Administration (FDA) granted accelerated approval to pembrolizumab in 2014 for the treatment of advanced melanoma. Following its initial approval, additional studies expanded its use to multiple cancer types. Even after approval, Phase IV post-marketing surveillance continues to monitor long-term safety and effectiveness in a larger patient population.

Section 3: Therapeutic Applications

Initially, pembrolizumab was approved only for the treatment of advanced melanoma. However, its use has expanded significantly and it is now approved for more than 40 cancer-related indications worldwide. Key therapeutic areas include:

Melanoma

The first cancer for which pembrolizumab received approval. It has shown substantial benefits in patients with advanced or metastatic disease.

Non-Small Cell Lung Cancer (NSCLC)

A major indication where the drug is particularly effective in patients with high PD-L1 expression, widely utilized as a first-line therapy.

Head and Neck Squamous Cell Carcinoma

Approved for recurrent or metastatic cases, improving survival outcomes and providing a vital alternative treatment option.

Urothelial Carcinoma

Utilized for patients with advanced bladder cancer whose disease progresses following platinum-based chemotherapy.

Triple-Negative Breast Cancer (TNBC)

An aggressive subtype of breast cancer. When combined with standard chemotherapy, pembrolizumab significantly improves clinical outcomes.

Cervical Cancer

Approved for PD-L1 positive recurrent or metastatic cases, offering an important additional therapeutic option.

Colorectal Cancer

Highly effective in tumors exhibiting high microsatellite instability (MSI-High) or mismatch repair deficiency (dMMR).

Tumor Mutational Burden-High (TMB-High)

A unique biomarker-based approval allowing its use across different solid tumors regardless of the anatomical site of origin.

Section 4: Market Impact

Pembrolizumab has had a major impact on both cancer treatment and the global pharmaceutical industry. Following FDA approval in 2014, the drug rapidly became one of the most successful immunotherapy products ever developed. Its approved indications have expanded to more than 40 cancer-related conditions, making it one of the most versatile oncology drugs available today.

By 2023, pembrolizumab generated approximately 25 billion US dollars in global sales, making it the highest-selling cancer drug worldwide. The success of Keytruda has contributed significantly to Merck's overall revenue and growth. The drug has also established immune checkpoint inhibitors as a major category of cancer therapeutics. Its success encouraged further research into PD-1 and PD-L1 inhibitors and accelerated innovation in cancer immunotherapy.

Despite these achievements, affordability remains a significant challenge, particularly in developing countries. In India, the cost of treatment remains high, limiting access for many patients. A single treatment cycle may cost a substantial amount, making long-term therapy difficult for some individuals. To address this challenge, several companies are developing biosimilar versions of pembrolizumab. These biosimilars may reduce treatment costs in the future and improve access to advanced cancer therapies.

Conclusion

Pembrolizumab has revolutionized cancer treatment by introducing immune checkpoint inhibition as a highly effective therapeutic strategy. By targeting the PD-1 pathway, it enhances the immune system's ability to recognize and destroy cancer cells. The drug has demonstrated proven efficacy across multiple cancers, including melanoma, lung cancer, bladder cancer, and several others. Although challenges related to cost and accessibility remain, ongoing research and biosimilar development may help overcome these barriers. Future studies are expected to further expand its indications and strengthen its role in modern oncology.

References

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